

Jayasri Suresh Vani

Software Engineer | Data Pipelines, Backend APIs & Applied AI/ML

+1 2604923581 | jayasri030520@gmail.com | linkedin.com/in/jayasri-suresh-vani-897879296 | github.com/Jayasri2021 | jayasri2021-96ad9.web.app/

SUMMARY

Software Engineer with 2+ years of experience across full-stack development, backend engineering, AI/ML, and data engineering. Built RAG pipelines, LLM evaluation systems, and full-stack applications, delivering measurable impact including a 45% reduction in reporting latency and 50% faster query performance. MS in Computer Science from Purdue University. Open to Software Engineer, Data Engineer, and AI/ML Engineer roles.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Java, TypeScript/JavaScript
- **Applied AI/ML:** RAG, LangChain, FAISS, LLM Evaluation, Prompt Engineering, NLP, Scikit-learn, Feature Engineering, LoRA Fine-Tuning
- **Backend & Web:** FastAPI, Flask, Spring Boot, Node.js, Next.js, React, Redux, REST APIs, GraphQL, MVC Architecture, JWT / RBAC
- **Databases Data Engineering:** PostgreSQL, MySQL, SQLite, Supabase, Prisma ORM, ETL, Data Pipelines, OLAP/OLTP
- **Cloud DevOps:** Azure, GCP (Cloud Run), AWS, Docker, Git, CI/CD
- **Testing Automation:** Cypress, Pytest, APScheduler, BeautifulSoup, HTTPX

EXPERIENCE

Generative AI Specialist | Community Dreams Foundation | Orlando, FL | Jul 2026 - Present

- Designed and deployed secure backend APIs integrating production applications with LLMs, enabling scalable AI-driven features.
- Optimized data pipelines and model inference workflows, reducing latency and cutting server infrastructure costs.
- Maintained core backend infrastructure on cloud platforms, ensuring high availability, data security, and efficient scaling.

Applied AI/ML Engineer | Raven AI | New York, NY | July 2025 - May 2026

- Engineered Python-based evaluation pipelines, converting 250+ unstructured medical dialogue datasets into structured behavioral scores.
- Benchmarked GPT-4o, Qwen3, and Llama using MAE, MSE, and R^2 metrics to drive model-selection decisions.
- Automated scoring and preprocessing workflows, accelerating prompt experimentation and reducing manual evaluation effort.

Lead Technical Support Engineer | Purdue University | Fort Wayne, IN | Dec 2024 - Apr 2026

- Led technical support operations across OS, networking, authentication, and AV systems for 500+ daily users.
- Standardized configuration and troubleshooting processes across 100+ systems, cutting resolution time through documented fixes.

Software Developer | Omega Binaries Pvt. Ltd | Chennai, TN | Dec 2023 - May 2024

- Developed a secure biometric authentication system using Java Swing, MVC architecture, and MySQL with encrypted credential storage.
- Implemented role-based access control and MFS100 fingerprint SDK integration for secure registration/login workflows.

Data Analyst & Machine Learning Engineer | Techvolt Software Pvt. Ltd. | Coimbatore, TN | Feb 2023 - Nov 2023

- Optimized SQL queries and reporting workflows, cutting reporting latency by 45% and building stakeholder-facing dashboards.
- Performed exploratory data analysis and cleaning on 10,000+ records, improving data quality for downstream modeling.
- Trained classical ML models (LR, XGBoost), achieving 85%+ predictive accuracy through cross-validation and hyperparameter tuning.

Applied Programming Instructor | CMS IT Groups | Mechanicsburg, PA | Jan 2022 - May 2024

- Delivered instruction in Python, algorithms, and database fundamentals to 100+ learners through project-based sessions.
- Mentored learners through 30+ coding exercises, developing 25+ technical resources to support hands-on learning.

EDUCATION

Purdue University - M.S., Computer Science | GPA: 3.9

Anna University - EEC College - B.Tech, Information Technology | GPA: 4.0

PROJECTS

CampusCrew | Student Services Platform

- Built a full-stack platform connecting PFW students to peer-offered services (tutoring, events) using Flask, React, and Redux.
- Designed REST APIs and a Supabase backend to manage service listings, bookings, and user roles.
- Implemented end-to-end test coverage with Cypress, improving release reliability across core user flows.

AuctionEase | Real-Time Auction Platform

- Developed a real-time bidding platform using Spring Boot, Java, and MVC architecture with React frontend.
- Designed RBAC and secure REST APIs to manage listings, bids, and transactions across user roles.
- Built ETL workflows to process auction and transaction data, reducing transaction latency by 40%.

NexusLMS-OP | Learning Management System

- Built a role-based LMS supporting 500+ users using Node.js, Next.js, Prisma, and PostgreSQL.
- Designed OLAP/OLTP data workflows to support both transactional operations and analytics dashboards.
- Optimized backend query retrieval time by 50% through schema and workflow improvements.

PharmaTrace | Pharmaceutical Data Warehouse

- Designed a data warehouse to track pharmaceutical export, batch quality, and cold-chain logistics data.
- Modeled complex domain relationships across batches, shipments, and quality checkpoints for reliable reporting.
- Built structured data pipelines to support compliance-driven analytics and traceability.

LumenDoc-QA | RAG Document Assistant

- Built a RAG pipeline with document ingestion, semantic chunking, and embedding generation for PDF and web content.
- Implemented grounded answer generation to reduce hallucination and improve source traceability.
- Tested retrieval quality across 50+ queries to evaluate answer relevance and grounding accuracy.

Gesture2Globe | ASL Recognitions & Translation

- Built a real-time computer vision pipeline for ASL recognition using hand-keypoint extraction and temporal modeling.
- Deployed a containerized inference backend on GCP Cloud Run, optimizing latency for real-time translation.
- Reduced inference latency by 50% through pipeline and deployment optimization.